Lecture 1 ERS312 Notes

How I became a Oceanographer (the Professor)

- Took bicycle ride in Egypt
- Snorkeling and saw jellyfish
- Directed studies towards oceans & marine geology
- As you go through sediment layers you are going through time to see how ocean system has behaved

What is Oceanography

- Study of the oceans
- Interdisciplinary (Geological: evolution of ocean basins, Physics: movement, Chemistry: properties of seawater, pH of seawater, Biology: looks at everything that lives in the ocean, History: history of human interaction with the ocean, Technology: need to make measurements

Why are the oceans important?

- Stores carbon dioxide
- Important food source
- High biodiversity
- Source of life
- Climate regulation (cool or heat the earth acts as a buffer)
- Important for transport
- Coastal areas are accessible
- Important for tourism
- Important for petroleum extraction

What problems do oceans currently face?

- Overfish oceans
- Coral reefs are deteriorating
- Ocean spills
- Dead zones due to nutrient pollution
- Rising temperatures in oceans (almost every part of ocean has warmed significantly) *Warming is more dramatic in the north (arctic) *sea ice disappearing in ocean *arctic amplification

Ocean Acidification

- Scale of time
- pH of ocean was stable at 8.2
- Ocean pH declining because burning of fossil fuels (increasing CO_2 increases CO_2 in ocean water more acidic)
- Over time decline in coral reefs

Marine pollution

- A lot of garbage in the ocean (accidental or tsunami can pick up garbage)
- Garbage patches in centre of Ocean Basins